Project Title: Parallel Sorter using Bitonic Sort in SystemVerilog

Due Date: Thursday May 15th, 2025

Objective:

The goal of this project is to design and implement a parallel sorter using the Bitonic sorting algorithm in SystemVerilog. The sorter should be able to handle a configurable number of elements efficiently. An optional bonus feature includes implementing the comparison phase in a serial manner while maintaining parallelism in the sorting operations.

Background:

Bitonic sort is a well-known parallel sorting algorithm that is highly suitable for hardware implementation due to its structured and deterministic behavior. It consists of multiple comparison and swap steps arranged in a log-based sequence. The use of comparators makes it ideal for FPGA-based sorting operations.

Bitonic sort is based on **Bitonic sequences**, which are sequences that first increase and then decrease (or vice versa). The sorting process consists of two main steps:

1. **Bitonic Sequence Construction**: The input data is arranged into Bitonic sequences through a series of merging steps.
2. **Bitonic Merging**: The Bitonic sequences are recursively broken down and sorted using comparators, ensuring the final sequence is fully sorted.

The algorithm operates in **log(n) stages**, where each stage performs parallel comparisons and swaps, making it well-suited for hardware implementations like FPGAs.

Project Requirements:

- Implement the Bitonic sort algorithm using SystemVerilog.
- The sorter should support a configurable number of input elements (e.g., 8, 16, or 32 elements).
- The design should be implemented and simulated on Quartus to verify the correctness of the sorting operation.

Optional Bonus Feature:

- Implement a version where the comparison operations are done serially, but the sorting operations remain parallel.

Deliverables:

- SystemVerilog source code for the sorter.
- Simulation waveforms showing correct sorting operation.
- A report detailing the design, implementation, and results.
- (Optional) Performance analysis comparing fully parallel vs. partially serial implementations.

Evaluation Criteria:

- Correctness of sorting implementation.
- Efficiency of hardware resource utilization.
- Modular and well-structured code.
- Quality of the report and analysis.
- Ability of answering questions during individual interviews.

Instructions on Project Submission:

Implement all designs on Intel Quartus Prime software.

Laboratory reports should be typed, and all graphs are computer generated. Arrange the report in the following order:

1. Objective: A brief statement of the purpose of the experiment.
2. Procedure: A brief summary of the general procedure without much detail. Show the schematics corresponding to your SystemVerilog. Include important connection diagrams, especially those which de ne the variables which were measured. Note that since you will already have done the work, you should use the past tense.
3. Results: The calculated results are to be presented in an organized manner. Show the simulation results for several random inputs as waveforms. Also, show the simulation results in a tabular form.
4. Discussion of results: The discussion is the most important item in the report. You should discuss the rationale behind your design and compare it with alternative designs.
5. Conclusion: This is a concise paragraph that summarizes your findings and points to an additional understanding of the subject covered in the experiment. The conclusion hereby should be of scientific nature and objective.

You are allowed to work in groups of three. There is a score on the presentation of the design. This score could vary from one student to another within a group depending on the performance of each student in answering questions.